# DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW
## SESSION 2024-2025: PERIODIC ASSESSMENT-I
## SUBJECT: MATHEMATICS (041)

SET–I

TIME: $1\frac{1}{2}$ Hrs. M.M.: 40

Dates :_____ Name : _____ Class/Sec IX_____

**General Instructions:**

**Read the following instructions carefully and follow them:**

- All questions are compulsory
- Q1 to Q8 carry one mark each.
- Q9 to Q11 are very short type questions and carry two marks each.
- Q12 to Q15 are short type questions and cary three marks each.
- Q16 to Q17 are long answer type questions and carry five marks each
- Q18 is case based question and is of four [1 + 1 + 2] marks.
- Internal choices are given

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## ➢ Multiple Choice Questions

1) The three rational numbers between 3 and 4 are:
   a) 5/2, 6/2, 7/2 b). 13/4, 14/4, 15/4 c). 12/7, 13/7, 14/7 d).11/4, 12/4, 13/4
2) One of the linear factors of $3x^2+8x+5$ is
   a) (x+1) b) (x-2) c). (x+2) d). (x-4)
3) If we multiply or divide both sides of a linear equation with the same non-zero number, then the solution of the linear equation:
   a) Remains the same b) Changes
   c) Changes in case of multiplication only d) Changes in case of division only
4) If x and y are both positive solutions of equation ax+by+c=0, always lie in the:
   a) . First quadrant b). Second quadrant c). Third quadrant d). Fourth quadrant
5) What is the class mark of the class interval 90-120?
   a). 90 b). 105 c). 115 d). 120
6) Ordinate of all points on the x-axis is
   a). -1 b). 0 c). 1 d). Any number

**Directions:** In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
(C) Assertion (A) is true but reason (R) is false.
(d) Assertion (A) is false but reason (R) is true.

7) **Assertion:** Rational number lying between two rational number x and y is $\frac{1}{2}(x+y)$.
   **Reason:** There is one rational number lying between two rational numbers.
8) **Assertion**: The point (2,2) is the solution of x + y = 4.
   **Reason**: Every point which satisfy the equation is the solution of equation.

## ➢ Very Short Answer Type Questions

9) Rationalize the denominator of $\frac{6}{\sqrt{5}-\sqrt{3}}$.
10) Express the linear equations $x - y/5 - 10 = 0$ in the form ax + by + c = 0 and indicate the values of a, b and c in each case:
11) Plot the points (1, 3), (– 1, – 1) and (– 2, – 3), and check whether they are collinear or not.

## ➢ Short Answer Type Questions

12) a) Write three solutions for each of the following equation: (i) $\pi x + y = 9$

**OR**

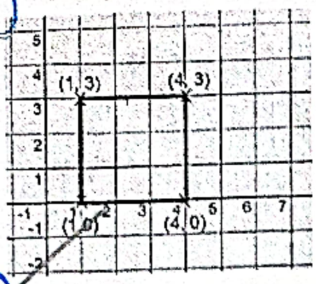
b) Find the value of k for 2a + 3b = k, if x = 2, y = 1 is a solution of the equation 2x + 3y = a and



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$x = -2, y = 3$ is a solution of $x - 3y = -b$.

13) Find the area of square given below in the figure




14) A family with a monthly income of Rs 20,000 had planned the following expenditures per month under various heads

Draw a bar graph for the data above.

| Heads | Expenditure (in thousand rupees) |
|---|---|
| Grocery | 4 |
| Rent | 5 |
| Education of children | 5 |
| Medicine | 2 |
| Fuel | 2 |
| Entertainment | 1 |
| Miscellaneous | 1 |

15) Represent the following on the number line

a) $\sqrt{18}$ OR b) $\sqrt{9.3}$

## Long Answer Type Questions

16) The following table gives the distribution of students of two sections according to the marks obtained by them:

| Section A | | Section B | |
|---|---|---|---|
| Marks | Frequency | Marks | Frequency |
| 0 - 10 | 3 | 0 - 10 | 5 |
| 10 - 20 | 9 | 10 - 20 | 19 |
| 20 - 30 | 17 | 20 - 30 | 15 |
| 30 - 40 | 12 | 30 - 40 | 10 |
| 40 - 50 | 9 | 40 - 50 | 1 |

Represent the marks of the students of both the sections on the same graph by two frequency polygons. From the two polygons compare the performance of the two sections.

17) Expand the following using suitable identity

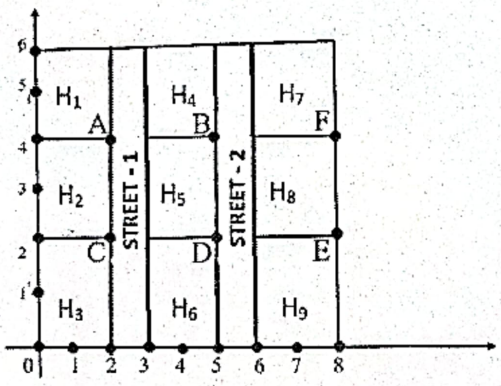
a) $(99)^3$ b) $(\sqrt{5}+\sqrt{3})^2-(\sqrt{5}-\sqrt{3})^2$

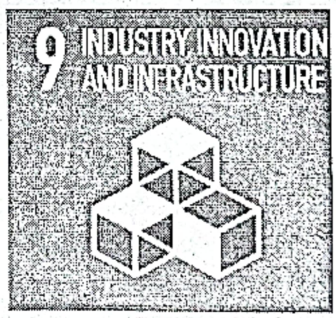
OR

Factorize the polynomial $P(x) = x^3 - 23x^2 + 142x - 120$.

## Case Study Based Questions

18) A contractor builds a society, there are three lane of row houses and two street are in between every two lanes. Every house is square in area of side 10 m and the width of both street is 5 m, if map of society is represented on cartesian plane graph and every block of graph is scaling by 5 m on graph, then answer the following:




a) Find the length of street -1

b) Find the distance between point C and E.

c) Find the coordinates of points A, B, D and E.



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